

STUDENT BOOK ANSWERS

Chapter 9 Reaction rates

Experiment 9.1: Observing rates of reaction

Analysis of results

- 1 Evidence of a chemical change; formation of gas, precipitate, decomposition, change of colour, change of temperature, light given off or change in odour
- 2 Students' answers may vary.
- 3 Students' answers may vary.

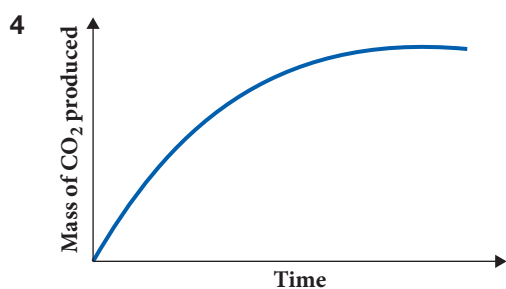
Discussion

- 1 Students' answers may vary.
- 2 Students' answers may vary.

Experiment 9.2: Measuring the rate of a reaction

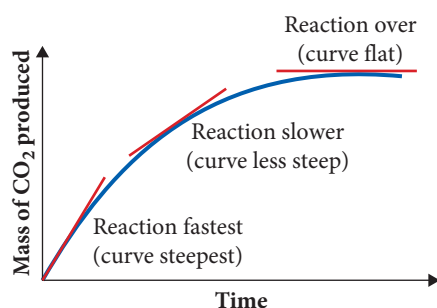
Analysis of results

- 1 The mass of the flask and its contents decreased over time. This was because carbon dioxide produced escaped from the flask.
- 2 The mass of the carbon dioxide produced is equal to the initial weight of the flask and contents minus the weight at the specified time.
- 3 Students' answers may vary. Based on 2.0 g, a total of 0.9 g should be lost in the time for the experiment.



- 5 The rate of reaction was fastest at the start, because that's when the gradient is the steepest.

- 6 Students' answers will vary for the specific time. At the end the gradient is the flattest, close to zero.
- 7 Students' answers will vary for the specific time. When the gradient was zero, at the end the reaction was complete. That is because no more carbon dioxide was being produced.
- 8 Students' answers will vary.
- 9 Students' answers may vary.



Discussion

- 1 Qualitative data is data that cannot be expressed as a number; e.g. presence of a particular element. Quantitative data can be expressed or measured as a number.
- 2 Students' answers may vary; e.g. by observing the rate of bubbling.
- 3 Students' answers may vary.
- 4 Students' answers may vary.
- 5 Students' answers may vary.

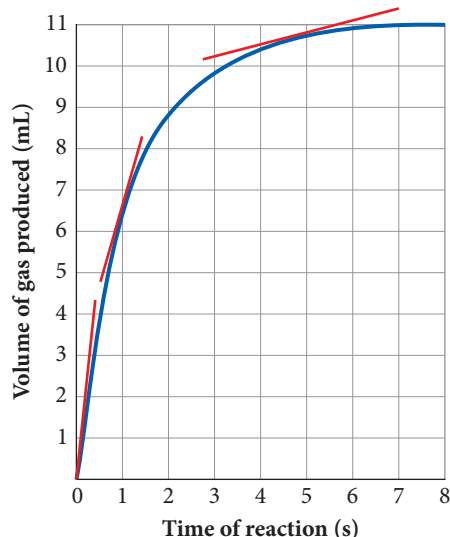
Conclusion

The rate is highest at the start of the reaction. It slows down as the reaction proceeds and becomes zero when the reaction stops.

Question set 9.1

- 1 The rate is the speed of a reaction; how fast or slow a reaction occurs.
- 2 The increase or decrease in mass, or production of a gas in a certain amount of time: The decrease in the amount of reactants or the increase in the amount of products in a certain time.
- 3 The rate is the fastest at the start.
- 4 Calculate the rate of a reaction by measuring the change in mass or volume over a certain amount of time. So if the mass of calcium carbonate decreases by 2 g in 2 minutes, the rate is 1 g min^{-1} or $\frac{1}{60} \text{ g per second}$.
- 5 The gradient of a graph of mass lost against time is equal to the rate of the reaction. When there is a large change in mass for a certain time the gradient is steep so then the rate is large. When there is little change in the mass for a certain time then the rate is slow, the gradient will be flatter. When there is no change in mass the gradient will be horizontal and there will be no visible rate of reaction.
- 6
$$\text{Rate} = \frac{20}{4 \times 60} = 0.08 \text{ g s}^{-1}$$

7


0: 11 mL CO₂ per s

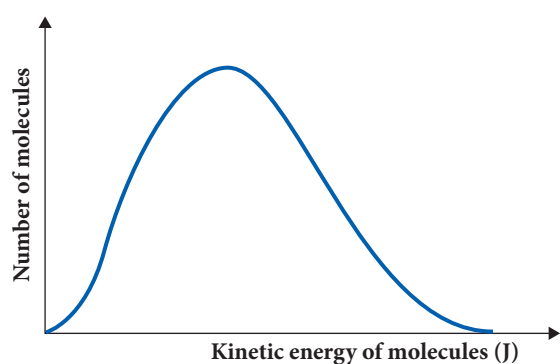
1: 3.5 mL CO₂ per s

5: 0.25 mL CO₂ per s

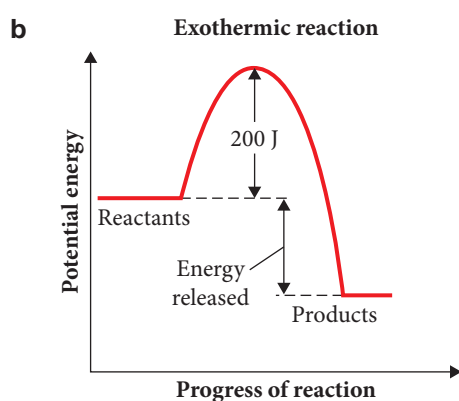
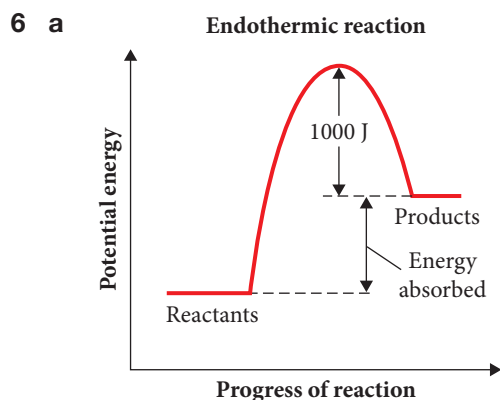
Question set 9.2

- 1 The collision theory states that, for a reaction to occur, the particles must collide with sufficient energy and the appropriate orientation to allow old bonds to be broken and new bonds to be formed.
- 2 a Activation energy is the energy required to start a reaction.
b The activated complex is the highly unstable structures formed when the reactants collide. It is the transitional structures formed when the reactants bonds are breaking and the molecules are starting to form the bonds in the products.

3

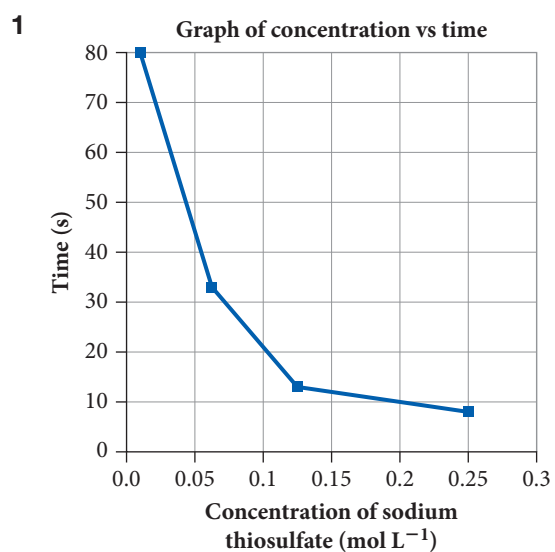


- 4 a The collision has to have enough energy to overcome activation energy of the reaction.
b The reactant molecules have to have the correct orientation so that if the reactant bonds are broken they are near the other atoms so that they can form the new bonds in the products.
- 5 Nothing. They will bounce off each other.



Experiment 9.3: Effect of concentration

Analysis of results



- 2** A faster reaction means that the cross will disappear faster.
- 3** As the concentration of sodium thiosulfate increased the time taken for the cross to disappear decreased.
- 4** As the concentration of sodium thiosulfate increases the rate of reaction increases.
- 5** Yes

- 6 Students' results may vary. Errors could be due to difficulty judging the time taken for the cross to disappear and different temperatures.

Discussion

- 1 As the concentration increases there are more collisions so the rate of reaction will increase.
- 2 Students' answers may vary. Difficulties include judging when the cross has disappeared, this would affect the time calculations.
- 3 Students' answers may vary. Use of a light meter to measure % light transmitted.

Conclusion

As the concentration increases there are more collisions so the rate of reaction will increase.

Taking it further

- 1 Students' answers will vary. Different range of concentrations could be tested.
- 2 Students' answers will vary. Different experiments could be used.

Experiment 9.4: Effect of surface area on rate of reaction

Analysis of results

- 1 Powder form
- 2 Powder form (The time taken for the bubbling to stop was the shortest for the powder form.)
- 3 Powder form, because it is crushed chips. For the same mass there is much greater surface area because inner particles are exposed.
- 4 As surface area increases, the rate of reaction increases.
- 5 Yes

Discussion

- 1 Students' answers may vary. The variables should be controlled and the same in all the trials, example, temperature at the start, concentration and type of acid, size of test tubes.
- 2 The exact masses of the calcium carbonate may not be identical in each trial. The reliability was improved by the repeating of the experiment. Judging the time for the bubbling to stop is also difficult and could make the results less accurate. The surface area of the lumps of calcium carbonate was not exactly the same.
- 3 The method could be improved with a better estimate for the time of reaction, for example; weigh the content of the test tube during the experiment. The pH of the sample could be measured as an indicator of rate.

Prepare blocks of calcium carbonate so that the surface area of the lumps are more consistent. The pH of the sample could be measured as an indicator of rate.

Conclusion

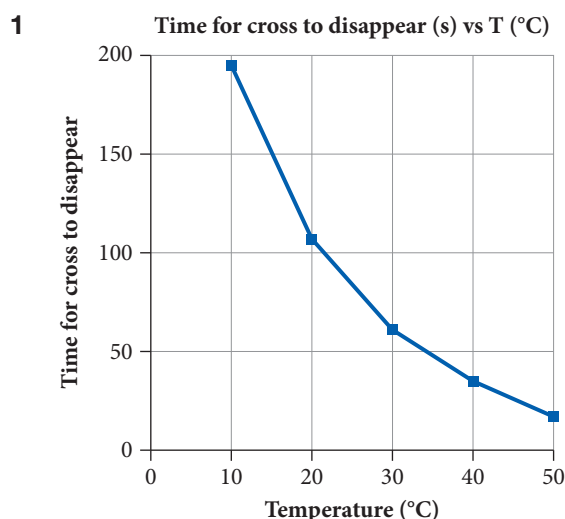
As the surface area increases, the rate of reaction increases.

Taking it further

Students' answers will vary. Examine a wider range of surface areas, examples blocks of calcium carbonate. Look at different reactions.

Experiment 9.5: Effect of temperature on rate of reaction

Analysis of results



2 The time taken for the cross to disappear decreased as the temperature increased.

3 The rate of reaction increases as temperature increases.

4 Yes

Discussion

1 Students' answers may vary. Difficulties include judging when the cross has disappeared; this would affect the time calculations. Temperature of the solution may not be uniformly consistent through the sample and the T measured may not be the true temperature.

2 Students' answers may vary.

3 Use of a light meter to measure % light transmitted. Allow more time for the temperature to stabilise before the reaction proceeds.

Conclusion

As temperature increases the rate of reaction increases.

Taking it further

Students' answers may vary. A wider range of temperatures could be tested.

Experiment 9.6: Effect of a catalyst on rate of reaction

Analysis of results

- 1 The time taken for the solution to decolourise was decreased.
- 2 Students should have found that the experiment supported the theory.

Discussion

- 1 Students' answers may vary. An alternative method of judging when the solution decolourised would make it more reliable. A method of adding all the ingredients and starting the stop watch could reduce errors. Freshness of the permanganate solutions could also have been an issue.

Conclusion

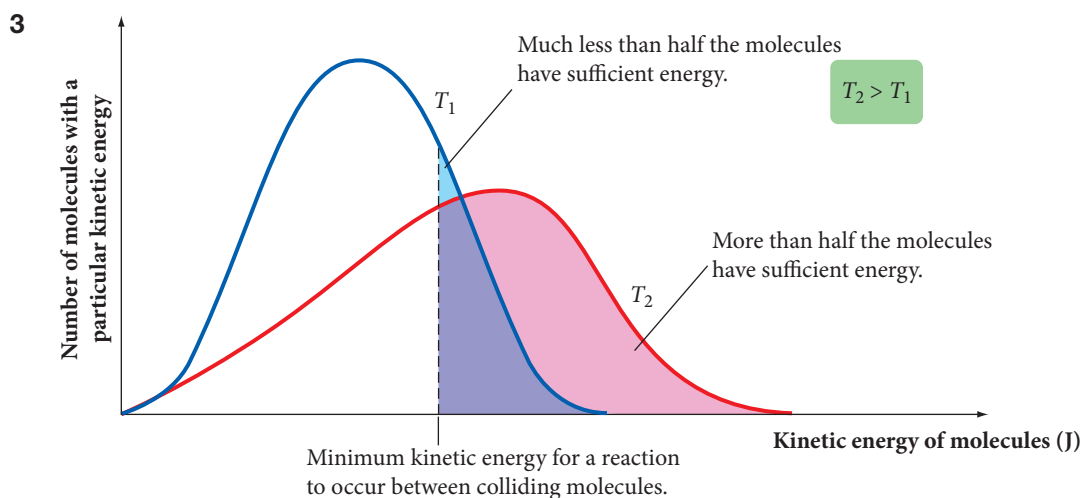
MnSO_4 will act as a catalyst for the reaction between oxalic acid and potassium permanganate, some reference to the actual observed difference in rate could also be included here.

Taking it further

- 1 Students' answers may vary. Trial different catalysts to see whether the effect is the same. Trial different reactions.
- 2 Students' answers may vary. Undertake the experiment varying other factors for the two test conditions of presence or absence of a catalyst.

Question set 9.3

- 1 Surface area, temperature, concentration and presence of a catalyst
- 2 A catalyst is a substance that affects the rate of a reaction without being consumed.



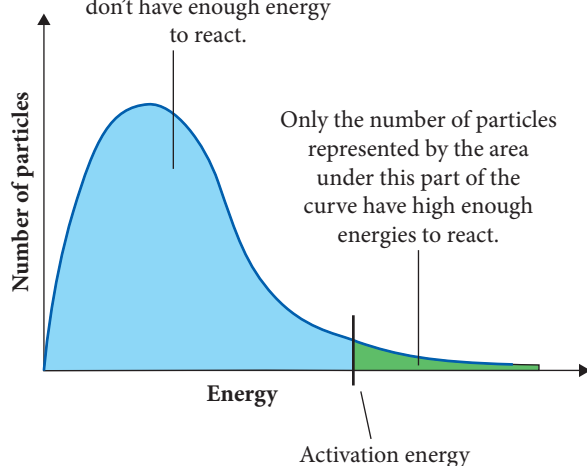
- 4 Crushing a lump of a solid exposes the inner particles to the reactant.
- 5 Substances at 10°C will have a lower average kinetic energy compared to substances at 40°C .
- 6 When the volume of a container is reduced the concentration of the gas has increased. This means the gas particles hit the side of the container more frequently and the pressure in the container increases. When the volume of a container is increased the concentration of the gas has decreased. This means the gas particles hit the side of the container less frequently and the pressure in the container decreases. This is how the pressure of a gas is equivalent to its concentration.

- 7
 - a The rate of reaction varies directly with the concentration. When the concentration (or pressure of a gas) increases there are more frequent collisions so the reaction rate will increase.
 - b The rate of reaction varies directly with the temperature increases. When the temperature increases a higher proportion of particles have enough energy to overcome activation energy so the reaction rate will increase. There are also more frequent collisions due to the increase in speed of the particles, this also increases the rate of reaction.
 - c The presence of a catalyst means that a lower activation energy level must be met for the reaction to proceed. This means that a higher proportion of particles have enough energy to react, so the rate of reaction increases.
 - d The rate of reaction varies directly with the surface area. With a higher surface area there are more frequent collisions so the rate of reaction increases.
- 8 Ben's tea would have been sweeter. As Ben stirred his tea the frequency of collisions between the water molecules and the sugar molecules increased. This meant that the rate of reaction for the sugar to dissolve has increased.
- 9 The higher the concentration, the higher the temperature and the higher surface area will produce the fastest rate. This occurs in test tube 5 as it has the highest value for all these variables.
- 10 With the understanding that particles have to collide with the correct energy and orientation we have a better understanding of why reactions occur at different rates. The different rates cannot be explained by only temperature or energy alone.

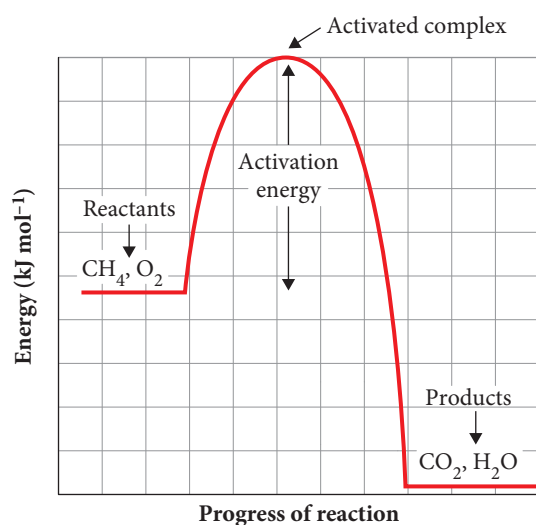
Chapter review questions

- 1
 - a Rate is change in a quantity (mass colour volume pH or concentration) over time.
 - b The total area of all the exposed surfaces
 - c The amount of solute dissolved in a given volume of solvent
 - d The force exerted by a gas or gases due to the collisions of the particles with the walls of a container; expressed as force per unit area
 - e A substance that changes the rate of a reaction without being consumed
- 2
 - Concentration: changing the concentration changes (increases or decreases) the number of particles involved in collisions
 - Surface area: the amount of surface area exposed to collisions is directly proportional to the rate of reaction
 - Temperature: change in temperature will change the average kinetic energy of particles and hence how many collisions have enough activation energy
 - Catalyst: catalysts provide an alternative pathway for the reaction which is easier and has lower activation energy
- 3 The quantity being measured such as mass, colour, volume, pH or concentration can be plotted against time. The gradient of the graph indicates the rate of reaction.

- 4 The number of particles represented by the area under this part of the curve don't have enough energy to react.

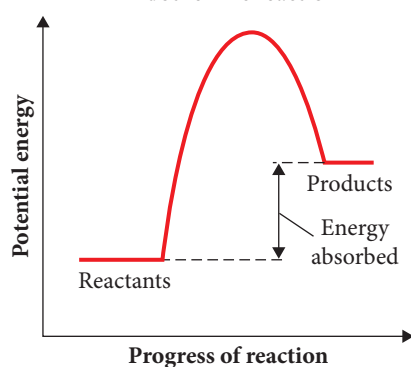


- 5



- 6 The particles must collide to break the bonds of the reactants before the bonds of the products can form. The energy required for the activation energy is provided by the collision.

- 7 Endothermic reaction

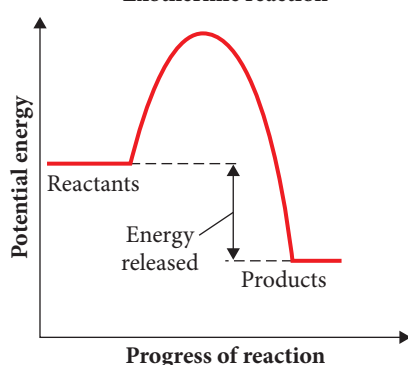


The energy of the reactants must be below that of the products. There must be a line drawn from the energy of the reactant to the products that has a higher energy than either. This represents the activation energy of the reaction. As the bonds are strong the activation energy must be large.

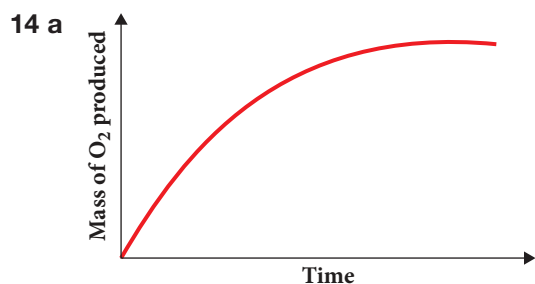
- 8 At the start of the reaction there is the highest concentration of the reactants. This means that the rate is the fastest as rate increases with concentration.
- 9 Catalysts increase the rate of reaction as they provide an alternative pathway for the reaction, with a lower activation energy. A greater proportion of the reactant particles will have enough energy to break their bonds on collision.
- 10 The activation complex has more energy than the reactants as energy is need to break the bonds of the reactants. It has more energy than the products as energy is lost when the bonds of the products are formed.

$$11 \text{ Rate} = \frac{\text{change in quantity 1}}{\text{time taken}} = \frac{\$525}{(6 \times 5\text{h})} = \$17.50\text{h}^{-1}$$

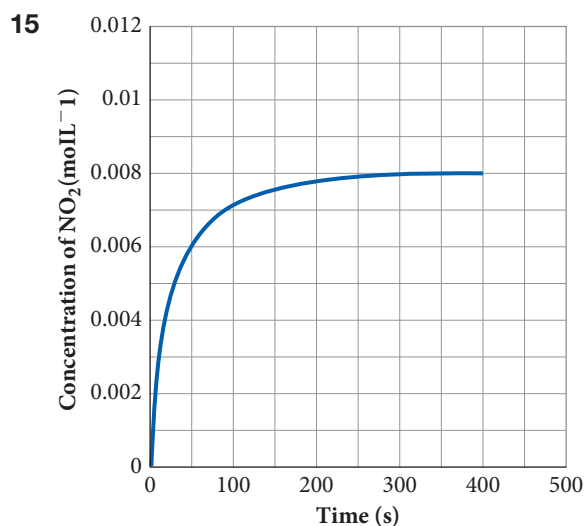
12 Exothermic reaction



- 13 In cold water the powder would take longer to dissolve as the rate of reaction with the water would be lower.



- b The manganese dioxide was a catalyst used to speed up the rate of reaction.
- c Cool the flask and hydrogen peroxide or dilute the hydrogen peroxide, or do not add the manganese dioxide. The volume of the gas should not alter. It may just take much longer to be produced.



16 Students' answers will vary. Reference could be made to the energy required in the two different paths, structures of the compounds along the path.

17 Students' answers will vary. They should make reference to the chance of the darts hitting the bull's eye and the chance for the particle to react.

18 Students' answers will vary.